

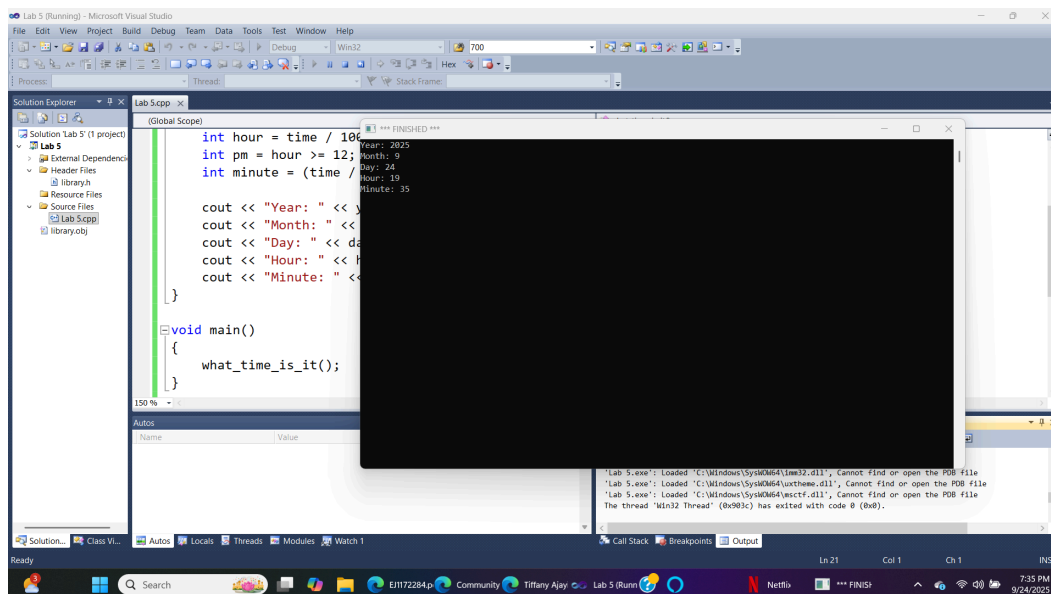
1. What time is it?

```
#include "library.h"
```

```
void what_time_is_it()
{
    int time = get_clock_time();
    int date = get_calendar_date();
    int year = date / 10000;
    int month = (date / 100) % 100;
    int day = date % 100;
    int hour = time / 10000;
    int pm = hour >= 12;
    int minute = (time / 100) % 100;

    cout << "Year: " << year << endl;
    cout << "Month: " << month << endl;
    cout << "Day: " << day << endl;
    cout << "Hour: " << hour << endl;
    cout << "Minute: " << minute << endl;
}
```

```
void main()
{
    what_time_is_it();
}
```



2. The Monroe Doctrine

```
#include "library.h"
```

```
void what_time_is_it()
```

```
{
```

```
    int time = get_clock_time();
```

```
    int date = get_calendar_date();
```

```
    int year = date / 10000;
```

```
    int month = (date / 100) % 100;
```

```
    int day = date % 100;
```

```
    int hour = time / 10000;
```

```
    int pm = hour >= 12;
```

```
        if (hour > 12)
```

```
            (hour = hour - 12);
```

```
        {
```

```
            if (hour >= 12) pm = true;
```

```
        }
```

```
    int minute = (time / 100) % 100;
```

```
    cout << "Date: " << year << "-" << month << "-" << day << endl;
```

```
    cout << "Time: " << hour << ":" << (minute < 10 ? "0" : "") << minute << (pm ? " p.m." : " a.m.") << endl;
```

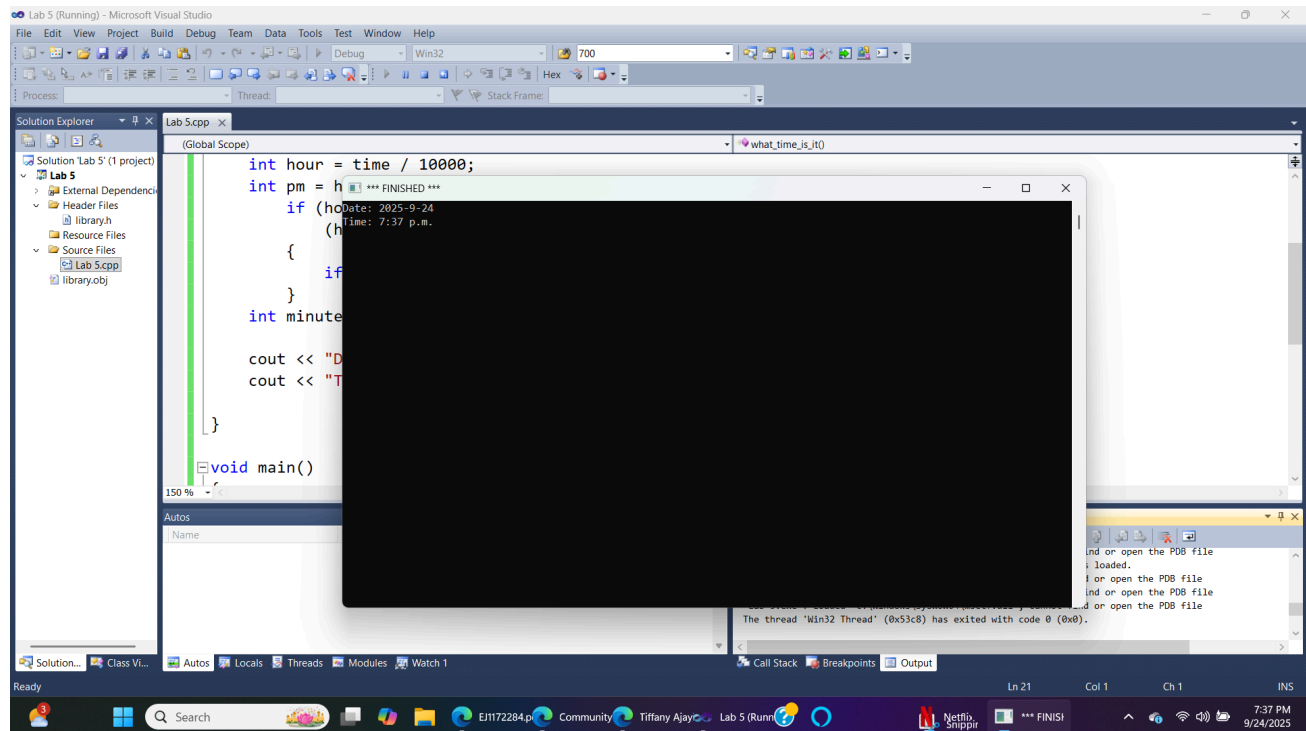
```
}
```

```
void main()
```

```
{
```

```
    what_time_is_it();
```

```
}
```



3. A Clock Face

```
#include "library.h"
```

```
void circle(double x, double y, double radius, const int steps)
```

```
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference/steps;
    double turn_angle = 360/steps;

    move_to(x + radius, y);
    set_heading_degrees(90);

    for (int i = 0; i < steps; i++)
    {
        draw_distance(step_length);
        turn_right_by_degrees(turn_angle);
    }
}
```

```
void clock_marks(int x, int y)
```

```
{
    double radius = 200;
    double steps = 360;

    for (int i = 0; i < 12; i++)
    {
        move_to(x,y);
        move_distance(radius - 15);
        draw_distance(15);
        move_distance(-(radius - 15));
        turn_right_by_degrees(30);
    }
}
```

```
void clock()
```

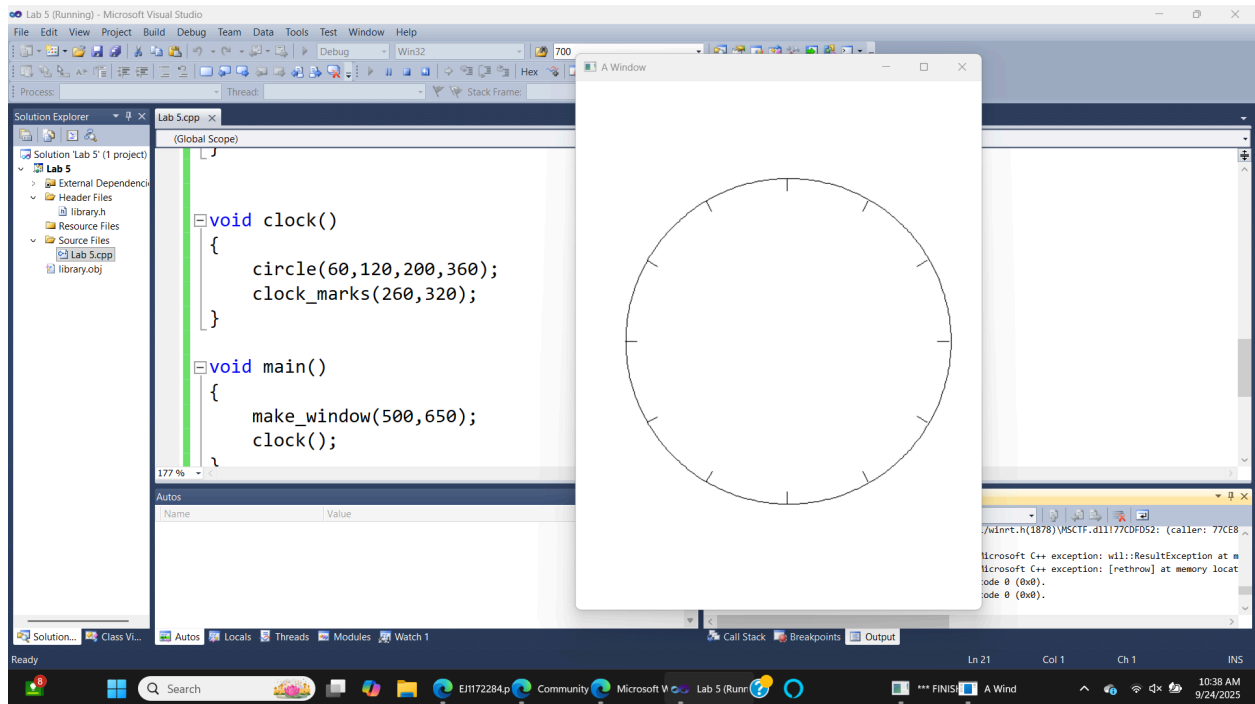
```
{
    circle(60,120,200,360);
}
```

```

        clock_marks(260,320);
    }

void main()
{
    make_window(500,650);
    clock();
}

```



4. A Whole Clock

```
#include "library.h"
```

```
void what_time_is_it()
```

```
{
    int time = get_clock_time();
    int date = get_calendar_date();
    int year = date / 10000;
    int month = (date / 100) % 100;
    int day = date % 100;
    int hour = time / 10000;
    int pm = hour >= 12;
    if (hour > 12)
        (hour = hour - 12);
    {
        if (hour >= 12) pm = true;
    }
    int minute = (time / 100) % 100;

    cout << "Date: " << year << "-" << month << "-" << day << endl;
    cout << "Time: " << hour << ":" << (minute < 10 ? "0" : "") << minute << (pm ? " p.m." : "
a.m.") << endl;
}
```

```
void circle(double x, double y, double radius, const int steps)
```

```
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference/steps;
    double turn_angle = 360/steps;

    move_to(x + radius, y);
    set_heading_degrees(90);

    for (int i = 0; i < steps; i++)
    {
        draw_distance(step_length);
        turn_right_by_degrees(turn_angle);
    }
}
```

```
}
```

```
void clock_marks(int x, int y)
```

```
{
```

```
    double radius = 200;
```

```
    double steps = 360;
```

```
    for (int i = 0; i < 12; i++)
```

```
    {
```

```
        move_to(x,y);
```

```
        move_distance(radius - 15);
```

```
        draw_distance(15);
```

```
        move_distance(-(radius - 15));
```

```
        turn_right_by_degrees(30);
```

```
    }
```

```
}
```

```
void clock_hands(double x, double y, double radius, int hour, int minute)
```

```
{
```

```
    double hour_angle = (hour % 12) * 30 + (minute * 0.5);
```

```
    double minute_angle = minute * 6;
```

```
    // hour hand
```

```
    move_to(x, y);
```

```
    set_heading_degrees(hour_angle);
```

```
    set_pen_width(2);
```

```
    draw_distance(radius * 0.5);
```

```
    // minute hand
```

```
    move_to(x, y);
```

```
    set_heading_degrees(minute_angle);
```

```
    set_pen_width(2);
```

```
    draw_distance(radius * 0.75);
```

```
}
```

```
void clock()
```

```
{
```

```
    int time = get_clock_time();
```

```

    int hour = time / 10000;
    int minute = (time / 100) % 100;

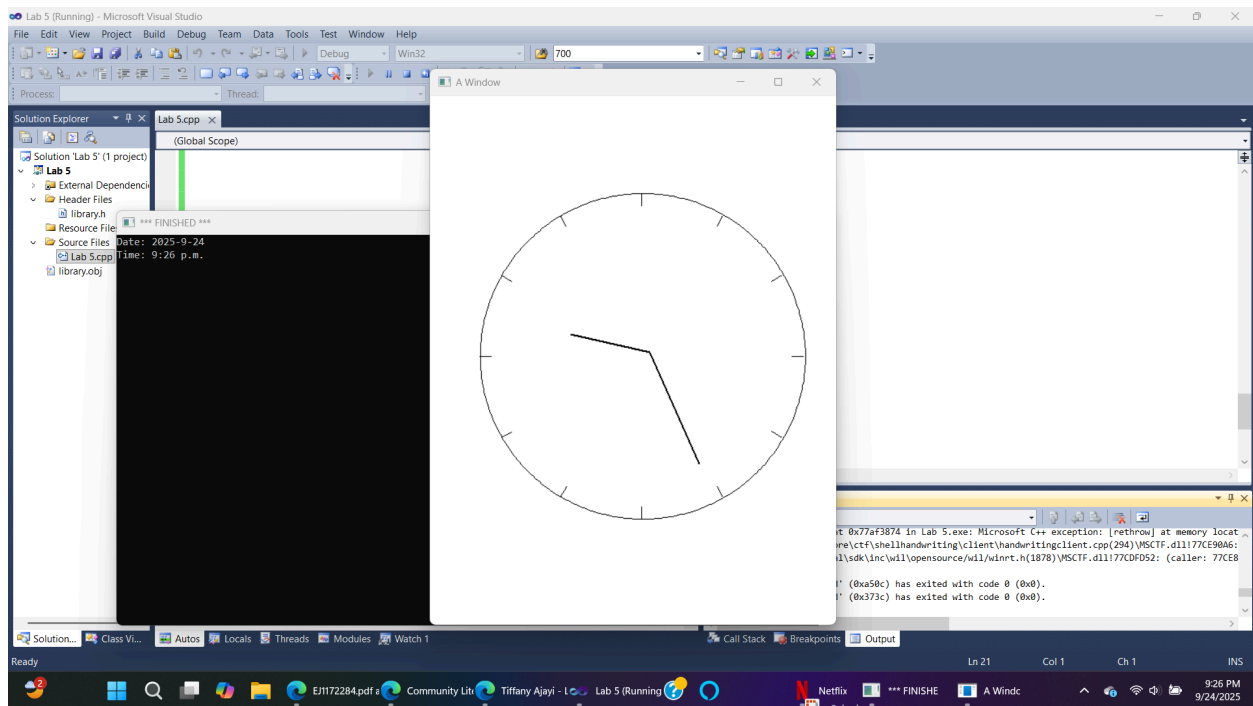
    circle(60,120,200,360);
    clock_marks(260,320);
    clock_hands(270,315,200,hour,minute);
}

```

```

void main()
{
    make_window(500,650);
    clock();
    what_time_is_it();
}

```



5. Add a Second Hand and Animate Your Clock

```
#include "library.h"
```

```
int time = get_clock_time();
int date = get_calendar_date();
int year = date / 10000;
int month = (date / 100) % 100;
int day = date % 100;
int hour = time / 10000;
int minute = (time / 100) % 100;
```

```
void what_time_is_it()
{
    int pm = hour >= 12;
    if (hour > 12)
        (hour = hour - 12);
    {
        if (hour >= 12) pm = true;
    }
    int seconds = time % 100;

    cout << "Date: " << year << "-" << month << "-" << day << endl;
    cout << "Time: " << hour << ":" << (minute < 10 ? "0" : "") << minute << (pm ? " p.m." : "
a.m.") << endl;
}
```

```
void circle(double x, double y, double radius, const int steps)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference/steps;
    double turn_angle = 360/steps;

    move_to(x + radius, y);
    set_heading_degrees(90);

    for (int i = 0; i < steps; i++)
    {
        draw_distance(step_length);
        turn_right_by_degrees(turn_angle);
    }
}
```

```
}
```

```
void clock_marks(int x, int y)
```

```
{
```

```
    double radius = 200;
```

```
    double steps = 360;
```

```
    for (int i = 0; i < 12; i++)
```

```
    {
```

```
        move_to(x,y);
```

```
        move_distance(radius - 15);
```

```
        draw_distance(15);
```

```
        move_distance(-(radius - 15));
```

```
        turn_right_by_degrees(30);
```

```
    }
```

```
}
```

```
void clock_hands(double x, double y, double radius, int hour, int minute, int seconds)
```

```
{
```

```
    double hour_angle = (hour % 12) * 30 + (minute * 0.5);
```

```
    double minute_angle = minute * 6;
```

```
    double seconds_angle = seconds * 6;
```

```
    // hour hand
```

```
    move_to(x, y);
```

```
    set_heading_degrees(hour_angle);
```

```
    set_pen_width(5);
```

```
    draw_distance(radius * 0.5);
```

```
    // minute hand
```

```
    move_to(x, y);
```

```
    set_heading_degrees(minute_angle);
```

```
    set_pen_width(5);
```

```
    draw_distance(radius * 0.75);
```

```
    // seconds hand
```

```
    move_to(x,y);
```

```
    set_heading_degrees(seconds_angle);
```

```
    set_pen_width(0.5);
```

```
    draw_distance(radius * 0.75);
```

```
}
```

```

void clock_face()
{
    circle(60,120,200,360);
    clock_marks(260,320);
}

void full_clock()
{
    int time = get_clock_time();
    int hour = time / 10000;
    int minute = (time / 100) % 100;
    int seconds = time % 100;

    clock_face();
    clock_hands(270,315,200,hour,minute,seconds);
}

void main()
{
    make_window(500,650);

    while (true)
    {
        wait(0.1);
        clear();
        full_clock();
        what_time_is_it();
    }
}

```

[Screen recording of animated working clock](#)

6. A Complete Product

```
#include "library.h"
```

```
int time = get_clock_time();
int date = get_calendar_date();
int year = date / 10000;
int month = (date / 100) % 100;
int day = date % 100;
int hour = time / 10000;
int minute = (time / 100) % 100;
```

```
void what_time_is_it()
```

```
{
    int time = get_clock_time();
    int date = get_calendar_date();
    int year = date / 10000;
    int month = (date / 100) % 100;
    int day = date % 100;
    int hour = time / 10000;
    int pm = hour >= 12;
    if (hour > 12)
        (hour = hour - 12);
    {
        if (hour >= 12) pm = true;
    }
    int minute = (time / 100) % 100;

    cout << "Date: " << year << "-" << month << "-" << day << endl;
    cout << "Time: " << hour << ":" << (minute < 10 ? "0" : "") << minute << (pm ? " p.m." : "
a.m.") << endl;
}
```

```
void circle(double x, double y, double radius, const int steps)
```

```
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference/steps;
    double turn_angle = 360/steps;

    move_to(x + radius, y);
    set_heading_degrees(90);
}
```

```

        for (int i = 0; i < steps; i++)
        {
            draw_distance(step_length);
            turn_right_by_degrees(turn_angle);
        }
    }

```

```

void clock_marks(int x, int y)

```

```

{
    double radius = 200;
    double steps = 360;

    for (int i = 0; i < 12; i++)
    {
        move_to(x,y);
        move_distance(radius - 15);
        draw_distance(15);
        move_distance(-(radius - 15));
        turn_right_by_degrees(30);
    }
}

```

```

void clock_hands(double x, double y, double radius, int hour, int minute, int seconds)

```

```

{
    double hour_angle = (hour % 12) * 30 + (minute * 0.5);
    double minute_angle = minute * 6;
    double seconds_angle = seconds * 6;

    // hour hand
    move_to(x, y);
    set_heading_degrees(hour_angle);
    set_pen_width(5);
    draw_distance(radius * 0.5);

    // minute hand
    move_to(x, y);
    set_heading_degrees(minute_angle);
    set_pen_width(5);
    draw_distance(radius * 0.75);

    // seconds hand
    move_to(x,y);

```

```
        set_heading_degrees(seconds_angle);
        set_pen_width(0.5);
        draw_distance(radius * 0.75);
    }
```

```
void clock_face()
{
    circle(60,120,200,360);
    clock_marks(260,320);
}
```

```
void full_clock()
{
    int time = get_clock_time();
    int hour = time / 10000;
    int minute = (time / 100) % 100;
    int seconds = time % 100;

    clock_face();
    clock_hands(270,315,200,hour,minute,seconds);
}
```

```
string month_name(int month)
{
    if (month == 1) return "January";
    if (month == 2) return "February";
    if (month == 3) return "March";
    if (month == 4) return "April";
    if (month == 5) return "May";
    if (month == 6) return "June";
    if (month == 7) return "July";
    if (month == 8) return "August";
    if (month == 9) return "September";
    if (month == 10) return "October";
    if (month == 11) return "November";
    if (month == 12) return "December";
}
```

```
string day_suffix(int day)
{
    if (day % 10 == 1 && day != 11) return "st";
    if (day % 10 == 2 && day != 12) return "nd";
```

```

    if (day % 10 == 3 && day != 13) return "rd";
    return "th";
}

void display_date_and_time(int text_size)
{
    int time = get_clock_time();
    int date = get_calendar_date();
    int year = date / 10000;
    int month = (date / 100) % 100;
    int day = date % 100;
    int hour = time / 10000;
    int minute = (time / 100) % 100;
    int pm = hour >= 12;
    if (hour > 12)
        (hour = hour - 12);
    {
        if (hour >= 12) pm = true;
    }

    set_font_size(text_size);
    move_to(20,60);
    write_string(day); write_string(day_suffix(day)); write_string(" ");
write_string(month_name(month)); write_string(" "); write_string(year);
    move_to(20,590);
    write_string(hour); write_string(":"); write_string(minute);
    write_string(pm? " p.m.":" a.m.");
}

void main()
{
    make_window(500,650);

    while (true)
    {
        wait(0.1);
        clear();
        full_clock();
        what_time_is_it();
        display_date_and_time(40);
    }
}

```

